



Antecedent configurations of organizational resilience in a major crisis: A strategic fit perspective

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ARTICLE INFO

Keywords:

Organizational resilience
Crisis responses
Dynamic capabilities
Institutional contexts
Crisis timeframes
Strategic fit

ABSTRACT

While research on organizational resilience has explored how organizations respond to adversity from either a capability-based or process-based perspective, the interplay between organizational capabilities and crisis responses remains underexplored. Drawing on the concept of strategic fit, we propose an integrated framework centered on the matching between key antecedents of organizational resilience—crisis responses and dynamic capabilities—which help firms navigate external shocks. We argue that the matching between specific capabilities and responses may vary depending on the institutional contexts and the crisis timeframes. Using the fsQCA approach and two datasets of listed firms from the U.S. and China during the COVID-19 crisis, we identify multiple configurations where crisis responses align with capability conditions to foster high resilience. These configurations differ based on the institutional contexts and the crisis timeframes. Overall, our study has significant implications for organizational resilience theory and crisis management practice.

1. Introduction

Crises such as terrorism, climate events, and pandemics significantly threaten organizational functioning and performance (James et al., 2011; Wenzel et al., 2020). Scholars have increasingly examined why some organizations exhibit greater resilience in such contexts (Conz & Magnani, 2020; Duchek, 2020). Two dominant perspectives have emerged: the capability view, which emphasizes resilience as a function of organizational capabilities like resource abundance and flexibility (Brito et al., 2022; Iftikhar et al., 2021); and the process-based view, which sees resilience as a dynamic process requiring deliberate crisis responses such as perception, resource restructuring, and innovation (Jiang et al., 2021; Williams & Shepherd, 2016; Yuan et al., 2022). While both views offer valuable insights, prior research remains fragmented, focusing on either capabilities or crisis responses, without integrating the two perspectives or clarifying their complex interdependencies (Duchek, 2020; Hillmann & Guenther, 2021; Khan et al., 2024; Linnenluecke, 2017).

To bridge this gap, we develop an integrative framework based on strategic fit (Venkatraman, 1989; Zajac et al., 2000), a perspective that highlights how organizational outcomes emerge from the alignment of capabilities, strategies, and environments. This approach implies that

organizational resilience may arise from multifaceted configurations of capabilities and crisis responses (Linnenluecke, 2017; Williams et al., 2017) across diverse institutional contexts and crisis timeframes (Jiang et al., 2021; Wang & Laufer, 2020). Despite growing recognition of these factors, prior studies have largely examined resilience within single institutional settings, overlooking cross-national differences (Napier et al., 2024). Additionally, these studies have provided limited empirical evidence on the evolution of resilience strategies over time.

Drawing upon strategic fit, we conceptualize organizational resilience as the outcome of dynamic capability-response configurations. Building on prior research, we focus on two key dynamic capabilities—strategic flexibility and organizational change capacity—and five critical crisis responses: crisis perception, persevering, retrenchment, corporate social responsibility (CSR) practices, and innovating (Ashraf et al., 2022; Wenzel et al., 2020). We argue that strategic flexibility aligns with resource-efficient approaches, as this capability enables organizations to reallocate and optimize resources (Zhou & Wu, 2010). Meanwhile, organizational change capacity matches resource-intensive strategies, as it could represent the resource capacity to manage large-scale transformations (Montreuil, 2023). We also expect these configurations may differ across market-led and state-led institutional frameworks due to variations in government roles and cultural

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<https://doi.org/10.1016/j.jbusres.2025.115646>

Received 13 March 2023; Received in revised form 7 August 2025; Accepted 7 August 2025

Available online 16 August 2025

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norms (Peng et al., 2008, 2009). Furthermore, as crises evolve, these configurations might change dynamically (Prayag et al., 2024).

Our empirical analysis examines 717 U.S.-listed and 890 Chinese-listed manufacturing firms that responded to Coronavirus Disease 2019 (COVID-19) from 2020 to 2022. These countries were chosen due to their comparable economic strength but distinct institutional contexts, with the U.S. representing a market-led framework and China a state-led framework (Fainshmidt et al., 2018; Hall & Soskice, 2001). We employ fuzzy-set qualitative comparative analysis (fsQCA) (Ragin, 2008) to identify configurations of capabilities and crisis responses across institutional contexts and crisis timeframes.

This study contributes to the organizational resilience literature in three ways. First, it advances an integrative framework based on strategic fit, showing how resilience arises from aligning organizational capabilities with crisis responses and the external environment. This offers a nuanced understanding of the complex interplay between capability- and process-based views of organizational resilience. Second, by comparing resilience configurations across U.S. and Chinese institutional contexts, it sheds light on how different institutional features affect resilience. Finally, it uncovers the evolving nature of resilience configurations over time, enriching the dynamic perspective of resilience research.

2. Literature review

Resilience has been widely studied across disciplines at individual, group, and system levels (Hillmann, 2021). In many fields, resilience is broadly defined as an entity's ability to withstand and adapt to disruptions at any level (Khan et al., 2024), while this study specifically examines its manifestation at the organizational level. With the increasing frequency and severity of global crises and persistent turbulence in the business environment, organizational resilience has become a central topic in management research. In the management domain, organizational resilience encompasses collective actions, strategic responses, and organizational capabilities that enable firms to navigate crises (Su & Junge, 2023)—distinct from individual or systemic resilience. Scholars have developed two distinct but complementary perspectives to understand organizational resilience: a capability-based view and a process-based view (Do et al., 2022; Duchek, 2020; Hillmann & Guenther, 2021).

Research grounded in a capability view considers organizational resilience as an inherent capability or *meta-capability* to navigate adversity (Anwar et al., 2023; Kahn et al., 2018). For instance, Ortiz-de-Mandojana and Bansal (2016) argue that organizational resilience arises from the ability to quickly process environmental signals and flexibly acquire and utilize resources. In line with this capability view, most studies examine the antecedents of organizational resilience by focusing on organizational and individual capability endowments (Yuan et al., 2022). Key antecedents include CEO traits (Sajko et al., 2021), organizational ambidexterity, and slack resources (Iborra et al., 2020; Woo et al., 2024). While these factors may not be crisis-specific, they can provide critical resources or enhance the ability of an organization to respond effectively to a crisis.

Other studies rooted in a process view suggest that organizational resilience is a dynamic process of active interaction between the organization and adversity (Williams et al., 2017). These studies extend beyond pre-existing and relatively static endowments, depicting how organizational actions evolve during crises to foster resilience. For example, Galkina et al. (2023) identify four progressive business model practices that enabled international firms to build resilience during COVID-19. This stream also differentiates between crisis phases and examines how organizational resilience unfolds over time. Conz and Magnani (2020) conceptualize organizational resilience as a dynamic process across pre-crisis, during-crisis, and post-crisis phases, defining distinct resilience elements for each phase.

Although prior studies offer valuable insights, scholars note that “the

study of resilience in the management and organizations' literature is fragmented” (Williams et al., 2017, p. 740), resulting in an incomplete understanding of its definition and formation (Duchek, 2020). They further argue that “organizational resilience is likely not just an ‘additive composite of individual’ resilience...but rather includes the interaction between an organization, its stakeholders, and the environment” (Lengnick-Hall et al., 2011, p. 245; Williams et al., 2017, p. 741). A few studies, recognizing its interactive and multifaceted nature, have explored how a single capability influences crisis response processes (e.g., Browder et al., 2024; Yuan et al., 2022) or vice versa (e.g., Smith et al., 2024). Yet the nuanced interaction mechanisms between capabilities and responses remain underexplored. To address this gap, we adopt a strategic fit perspective and develop an integrative framework to examine how capabilities and responses jointly shape organizational resilience.

3. Theoretical framework

3.1. A strategic fit perspective of organizational resilience

Strategic fit refers to the matching or alignment between an organization's strategy and its environmental or organizational contingencies, with positive performance implications (Venkatraman, 1989; Zajac et al., 2000). Building on contingency theory's notion of situational adaptability, this theoretical lens explains how organizations adapt to changing internal and external conditions, offering specific guidance on achieving and maintaining strategic alignment through dynamic matching mechanisms (Fainshmidt et al., 2019; Zajac et al., 2000). These mechanisms are crucial for understanding the multifaceted and evolving nature of organizational resilience. For example, during COVID-19, many retailers quickly shifted to e-commerce, while some apparel firms reconfigured production to make protective equipment, both reflecting strategic fit in crisis response.

Based on the concept of strategic fit, we argue that organizational resilience as a multifaceted and complex construct should be defined by integrating capabilities—particularly dynamic capabilities—with crisis responses, along with specific environments. Following Williams et al. (2017), we define organizational resilience as *both a capacity and a process through which firms build and leverage their capability endowments to proactively interact with the environment amid evolving adversity*. This perspective captures both the static and dynamic aspects of resilience, highlighting its proactive, adaptive, and evolving nature. Empirically, resilience can be reflected in reduced losses, faster recovery, and the implementation of active and effective actions.

3.2. Antecedents and their configurations of organizational resilience

Following prior research, we identify key antecedents of organizational resilience and examine their alignment and configuration. From the process-based view, we consider crisis perception and response strategies (Williams et al., 2017). From the capability-based view, we focus on two dynamic capabilities: strategic flexibility (Jiang et al., 2019) and organizational change capacity (Duchek, 2020), which provide the agility and adjustable resources to respond effectively to crises (Mackay et al., 2020; Montreuil, 2023). Crisis perception is the process of identifying and understanding a crisis (Billings et al., 1980). It helps firms recognize early warning signs and gather information to understand the situation, enabling them to formulate effective response strategies that enhance organizational resilience (Ghobadian et al., 2022; Kim, 2021). Besides crisis perception, adopting suitable crisis response strategies is essential to withstand adversity and develop resilience.

Grounded in previous studies, we focus on four key crisis response strategies particularly relevant in our crisis context: persevering, retrenchment, CSR practices, and innovating (Ashraf et al., 2022; Wenzel et al., 2020). *Persevering* describes measures to sustain a firm's business activities in response to a crisis (Wenzel et al., 2020). By

preserving core resources and maintaining essential operations, persevering helps organizations weather adverse conditions and build resilience (Li et al., 2022). Li and Tallman (2011) found that firms need to persevere to reap long-term benefits in times of crisis. *Retrenchment* refers to “reductions in costs, assets, products, product lines, and overhead” (Pearce & Robbins, 1993, p. 614). Through efficient resource allocation and cost control, retrenchment enables organizations to maintain operations and improve efficiency during crises, contributing to organizational resilience (Tao et al., 2020). *CSR practices* refer to the actions taken by firms to minimize negative social and environmental impacts while promoting the public good (Sheehy, 2015). CSR practices enhance organizational resilience by strengthening stakeholder relationships and securing external support during crises (Ashraf et al., 2022). Magrizos et al. (2021) found that Greek firms’ CSR practices improved their financial performance during economic crises. *Innovating* refers to the “realization of strategic renewal in response to crisis” (Wenzel et al., 2020, p. 11). By exploring emerging opportunities and developing new solutions, innovating helps organizations adapt to crisis conditions and improve their resilience (Roper & Turner, 2020; Wan & Yiu, 2009). Naidoo (2010) noted that innovation helps firms develop competitive advantages during economic crises, ensuring their survival.

While crisis response strategies play a crucial role in organizational resilience, their effectiveness is likely contingent upon firms’ dynamic capabilities—the ability to adapt, integrate, and reconfigure resources and competencies in response to rapidly changing environments (Teece et al., 1997). We focus on two key dynamic capabilities: strategic flexibility and organizational change capacity, because they could represent firms’ resource allocation capability and resource capacity to cope with a crisis, respectively (Montreuil, 2023; Sanchez, 1995). *Strategic flexibility* is defined as “the ability of a firm to reallocate and reconfigure its organizational resources, processes, and strategies to deal with environmental changes” (Zhou & Wu, 2010, p. 549). It enhances organizational resilience by enabling firms to swiftly identify threats and opportunities and efficiently redeploy resources toward core functions or emerging opportunities (Nadkarni & Narayanan, 2007; Zhou & Wu, 2010). *Organizational change capacity*, the capability to initiate and achieve various types of changes continuously (Heckmann et al., 2016), reflects an organization’s key resources and its capacity to manage change (Montreuil, 2023). It strengthens organizational resilience by providing resource support for large-scale changes and leveraging change experience to navigate crises effectively (Klarner & Raisch, 2013; Meyer & Stensaker, 2006).

According to the concept of strategic fit, effective alignment between specific crisis responses and appropriate capabilities could produce configurations that contribute to organizational resilience (Fainshmidt et al., 2019; Prayag et al., 2024). On the one hand, crisis perception, as an immediate response to a crisis, might need to align with dynamic capabilities, which enable firms to sense early signs of a crisis quickly and effectively (Jiang et al., 2021). On the other hand, the implementation of crisis response strategies, as planned reactions to a crisis, may require different capability supports. Specifically, the long-term persevering strategy might need the support of organizational change capacity to ensure continued resource availability (Wenzel et al., 2020). Retrenchment is likely to align with strategic flexibility, as it could help firms reduce costs and enhance value through efficient resource reallocation and operational adjustments (Lee & Makhija, 2009). Additionally, resource-intensive strategies such as CSR practices and innovating may need to match with organizational change capacity. This is because change capacity represents firms’ accumulated resources and experience in managing large-scale changes (Montreuil, 2023). Such capacity provides the necessary foundation of resources and competencies for implementing comprehensive CSR initiatives and pursuing innovative transformations (Dolata, 2009; Ingham & Havard, 2017).

3.3. Institutional context of configurations

From a strategic fit perspective, the configurations of capabilities and crisis responses should align with the institutional contexts. The institutional theory suggests that organizational behaviors are shaped by surrounding formal (i.e., laws and regulations) and informal institutions (i.e., cultures and norms), as institutions define “the rules of the game” (North, 1990; Peng et al., 2008). According to Fainshmidt et al. (2018) and Peng et al. (2008), there are two distinct institutional frameworks: market-led (i.e., the U.S.) and state-led (i.e., China). Their differences in formal and informal institutions may influence firms’ approaches to crisis response and resilience building (Wang & Laufer, 2020).

Market-led environments (U.S.) feature regulatory stability and predictable legal frameworks (Hall & Soskice, 2001) that may affect capability-response configurations in two main ways: by (1) providing a more stable institutional environment that enables firms to rely more on market mechanisms, such as financial retrenchment combined with strategic flexibility; and (2) framing CSR as discretionary rather than obligatory. The individualist cultural orientation could further legitimize autonomous decision-making and efficiency-driven responses over collective welfare considerations (Sims, 2009).

Conversely, state-led environments (China) are characterized by active government intervention (Peng et al., 2008), which may: (1) necessitate stronger dynamic capabilities to navigate policy uncertainties; and (2) limit firms’ use of retrenchment while elevating CSR as a core strategic element to maintain legitimacy. These formal institutional pressures might be reinforced by collectivist culture that prioritizes relationship maintenance and social harmony during crises (Shafer et al., 2007).

Empirical studies suggest that such institutional differences affect firms’ crisis responses. Eng et al. (2019) found that compared to Chinese firms, U.S. firms adopted more market-based adjustments like real earnings management following the 2008 financial crisis. Therefore, we expect that the way firms configure their crisis responses and dynamic capabilities to achieve resilience may differ across institutional contexts.

3.4. Time frame of configurations

As crises evolve, organizations need to dynamically adjust their capabilities and responses to adapt to the changing external environment (Browder et al., 2024; Conz & Magnani, 2020). As Tasic et al. (2020, p. 714) suggest, “organizational resilience emerges from interactions among variables at different levels that evolve over time, and thus its analysis requires a multilevel and dynamic perspective”. Therefore, it is necessary to match organizational capabilities and crisis responses with different periods of a crisis (Jiang et al., 2019; Linnenluecke et al., 2012).

In the short-term period, when external shocks create high uncertainty, organizations need to quickly detect threats and maintain operational stability (Duchek, 2020; Jiang et al., 2021). Crisis perception and perseverance are particularly crucial during this phase, as firms focus on immediate survival and uncertainty reduction (Linnenluecke et al., 2012). In the long-term period, as the crisis persists, organizations shift from mere survival to strategic adaptation, with the importance of initial crisis responses gradually diminishing (Jiang et al., 2019). During this period, organizations may increasingly emphasize sustainable practices and strategic adjustments, such as CSR and innovation, to build long-term competitive advantages (Prayag et al., 2024). Thus, we expect firms within specific institutional contexts to align different crisis responses with dynamic capabilities across various timeframes to build resilience. This implies that antecedent configurations of organizational resilience may evolve over time.

In summary, we argue that strategic fit is critical at multiple levels: internally, where crisis responses must align with dynamic capabilities; externally, where responses need to match institutional contexts; and temporally, where configurations evolve across different crisis periods.

This multi-level perspective on strategic fit forms the theoretical foundation of our framework, as illustrated in Fig. 1.

4. Methods

We use fsQCA, a configurational approach based on set logic, to test our framework (Ragin, 2008). Unlike regression analysis, which focuses on linear relationships, fsQCA excels in capturing configurational complexity by explaining how multiple conditions combine in different ways to produce an outcome (Furnari et al., 2021). This approach has been increasingly used in crisis and resilience studies (e.g., Jiang et al., 2024; Li et al., 2022).

4.1. Data and sample

We focus on the COVID-19 pandemic as the crisis context, which provides a unique opportunity to examine how firms build resilience and compare it across countries. Our sample includes U.S. and Chinese-listed manufacturing firms producing consumer products during the pandemic. We selected these firms for two reasons: First, the U.S. and China represent market-led and state-led institutional frameworks with comparable economic strengths. Second, consumer product manufacturing was severely affected by COVID-19 on both the demand and supply sides. Although sectors like tourism and other services were also impacted, their offline nature limits the effectiveness of strategic responses in facilitating demand recovery (Fotiadis et al., 2021).

Consistent with prior research (Benton et al., 2022), we identify 2020 as the onset of the COVID-19 outbreak to observe firms' crisis responses and resilience performance. Our first step was to specify the industries of interest. For U.S. firms, we selected 13 industries based on two-digit Standard Industrial Classification (SIC) codes and 9 industries based on three-digit SIC codes. For Chinese firms, we chose 25 industries based on the China Securities Regulatory Commission (CSRC) classification standards. These industries are closely linked to consumers downstream, such as the food and clothing sectors. A detailed list is presented in the Appendix. We then searched the COMPUSTAT (for U.S. firms) and CSMAR (for Chinese firms) databases for firms in these industries from 2020 to 2022. After excluding foreign firms listed in the U.S. and those listed for less than five years, we obtained an initial sample of 1,210 U.S. firms and 1,061 Chinese firms.

Our data was compiled from multiple sources. Crisis perception and response strategies data were collected from firms' 2020 periodic reports using text analysis methods. Data on organizational resilience and dynamic capabilities came from CSMAR and WIND (Chinese firms) and COMPUSTAT, SDC Platinum, BoardEx, and CRSP (U.S. firms). After removing missing values and outliers (above 99th and below 1st percentile) and merging by ticker symbols, we obtained a final sample of 717 U.S. firms and 890 Chinese firms. The smaller U.S. sample is mainly due to greater data fragmentation and missing values across multiple

databases. Robustness checks using industry-stratified subsamples confirm that our findings are stable despite this difference in sample size (see Appendix).

4.2. Measures

Crisis perception and crisis response strategies. Measuring corporate crisis perception and crisis response strategies during the COVID-19 pandemic is a challenging task. In this study, we employed a text analysis method that combines manual identification and machine learning to construct our measurements.

First, we obtained firms' periodic reports (i.e., 10-K or 10-Q filings for U.S. firms and semi-annual or annual reports for Chinese firms) with periods ended (or closest to) June 30, 2020, from the official websites of securities regulatory authorities in the U.S. and China using Python. We extracted sentences containing "COVID" or "pandemic," segmented them by periods, and saved them in TXT format as raw text data. Reports from this period were chosen because most Chinese firms did not disclose COVID-related information earlier, and by June 2020, the pandemic had largely stabilized in both countries, marking the end of the initial phase of crisis response.

Second, we extensively reviewed the raw text and consulted prior studies on corporate crisis responses to COVID-19 (e.g., Ghobadian et al., 2022; Raj et al., 2022; Wang et al., 2020; Wenzel et al., 2020) to build a keyword dictionary. Table 1 provides an excerpt from this dictionary, which includes keywords and keyword phrases representing crisis perception and four types of crisis response strategies.

Finally, based on this dictionary, we performed keyword matching on the raw text data (Niu et al., 2025). We used the spaCy library to preprocess the text, including lemmatization, tokenization, and removal of stop words, while retaining punctuation for proper sentence segmentation. For each firm's sentences, if any keyword corresponding to a specific crisis response was found, that response was assigned a value of 1; otherwise, 0. Keywords preceded by negations and those referring to third parties (e.g., government agencies and suppliers) were excluded.

Strategic flexibility. Following Nadkarni and Narayanan (2007), we measured strategic flexibility as the industry-adjusted coefficient of variation of capital intensity (capital expenditures/total revenue), R&D intensity (R&D expenditures/total revenue), and advertising intensity (advertising expenditures/total revenue). We calculated these coefficients annually from 2015 to 2019, averaged them, and adjusted for industry effects by subtracting the industry median. Higher coefficients indicate lower flexibility.

Organizational change capacity. Based on Morales-Raya and Bansal (2015), we measured organizational change capacity through firms' speed of change, a core characteristic of change capacity (Amis et al., 2004) particularly relevant during the COVID-19 crisis (Ahlstrom & Wang, 2021). Operationally, we calculated and standardized three indicators from 2015 to 2019: the number of M&As, average executive

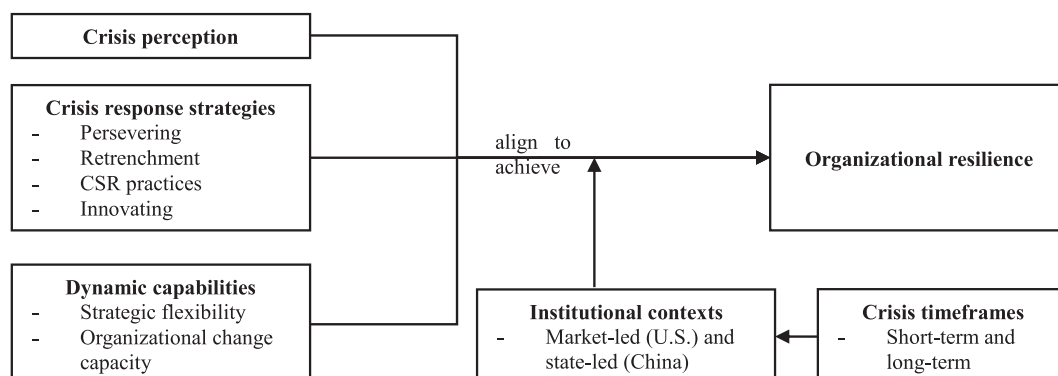


Fig. 1. Theoretical framework.

Table 1
Keyword dictionary used for measurement.

Conditions	Keywords (or keyword phrases)
Crisis perception	Monitor; assess; evaluate; contingency; emergency; proactive; active; aggressive; taskforce; establish, committee; leadership, team; crisis, team; crisis, management; cross-functional; prevention, control; formulate, plan.
Persevering	maintain, operation; resume, operation; restart, operation; return, work; reopen, office; continue, operate; uninterrupted, operation; business, continuity; ensure, manufacturing; remote, work; strengthen, liquidity; borrow, loan; amend, credit; ensure, cash; preserve, liquidity; negotiate, supplier; government, relief; defer, payment; health, safety, protocol; social, distancing; temperature, check; clean, sanitize; protective, equipment; solidify, relationship; remain, contact; enter, agreement; continue, invest; increase, sale; increase, manufacturing.
Retrenchment	layoff; furlough; control, cost; save, cash; reduce, expense; reduce, inventory; reduce, workforce; manage, inventory; manage, liquidity; restructure; improve, efficiency; streamline; optimize; limit, travel; limit, expend; suspend, interaction; suspend, dividend; suspend, operation; close, facility; idle, facility; close, factory; cancel, order; delay, commercialization; hiring, freeze.
CSR practices	employee, safety; employee, health; employee, bonus; employee, incentive; employee, benefit; protect, employee; increase, wage; support, community; support, customer; produce, mask; produce, medicine; develop, medicine; develop, treatment; develop, vaccine; develop, diagnostic; develop, drug; develop, antibody; donate; charitable, contribution; relief, effort.
Innovating	new, product; new, business; new, market; channel, expansion; market, development; virtual, service; contactless, service; contactless, delivery; online, channel; digital, channel; ecommerce, channel; invest, ecommerce; shift, ecommerce; digital, marketing; influencer, marketing; online, marketing; live streaming, marketing; online, promotion.

Notes: 1. The table presents an excerpt of the keywords representing crisis perception and the four crisis response strategies, with different keywords separated by semicolons. 2. The dictionary includes keyword phrases, such as “control, cost,” which represent the concept of “cost control.” In each keyword phrase, different words are separated by commas. If different words from a keyword phrase are detected within a single sentence, it is considered a match for that keyword phrase.

team tenure (reversed), and average annual stock trading volume. The sum of these standardized values was industry-adjusted, with higher values indicating greater change capacity.

Organizational resilience. Consistent with Li et al. (2023), we evaluated organizational resilience by examining changes in operating revenue per unit production cost (ORPPC) before and after the crisis. Specifically, organizational resilience was measured as the industry-adjusted ratio of a firm’s ORPPC in Q2 of 2020 (post-crisis) to its ORPPC at the end of 2019 (pre-crisis). This measure captures a firm’s operational performance during the crisis, with higher values indicating lower losses and greater recovery, thereby reflecting stronger resilience. To account for the lagged effects of crisis responses on organizational resilience, we also used the industry-adjusted ratio based on Q3 of 2020 (post-crisis) for robustness checks, and the results are largely robust (see Appendix). Table 2 summarizes all measures and data sources discussed above.

4.3. Analytical technique

The analysis in this study was conducted using the QCA package in the R programming language, following the fsQCA guidelines by Pappas and Woodside (2021). The first step was data calibration, where the raw values of continuous conditions and outcome (strategic flexibility, organizational change capacity, and organizational resilience) were converted into fuzzy set membership scores. We used Ragin’s (2008) direct calibration method, setting the 95th, 50th, and 5th percentiles of these conditions and outcome as the thresholds for full membership, crossover, and full non-membership, respectively (with reversed calibration thresholds for strategic flexibility). Crisis perception and

Table 2
Summary of measures and data sources for conditions and outcome.

Conditions/Outcome	Measures	Data sources
Crisis perception and response strategies	Text analysis of COVID-19-related sentences in firms’ periodic reports with periods ended June 30, 2020, using keyword matching based on an established dictionary (see Table 1). Binary coding (1 if keyword present, 0 if absent).	U.S. and Chinese securities regulatory authorities’ websites
Strategic flexibility	Industry-adjusted coefficient of variation of: Calculated annually from 2015 to 2019 and averaged. - Capital intensity (capital expenditures/total revenue) - R&D intensity (R&D expenditures/total revenue) - Advertising intensity (advertising expenditures/total revenue)	CSMAR and WIND (Chinese firms) COMPUSTAT (U.S. firms)
Organizational change capacity	Industry-adjusted sum of standardized values (2015–2019) of: - Number of M&As - Average executive team tenure (reversed) - Average annual stock trading volume	CSMAR and WIND (Chinese firms) SDC Platinum, BoardEx, and CRSP (U.S. firms)
Organizational resilience	Industry-adjusted ratio of ORPPC in Q2 2020 to ORPPC at the end of 2019.	CSMAR and WIND (Chinese firms) COMPUSTAT (U.S. firms)

response strategies were coded as crisp sets (0 or 1) and did not require calibration. Table 3 presents the calibration thresholds and descriptive statistics for each condition and outcome.

For the sufficiency analysis, we applied thresholds for frequency, consistency, and proportional reduction in inconsistency (PRI) to minimize the truth table. Following Ragin’s (2008) recommendations, the frequency thresholds for U.S. and Chinese firms were set at 6 and 10, respectively, to retain approximately 80 % of the cases. Consistency thresholds were set at 0.800, consistent with common practice. Given the lower PRI levels for U.S. firms, we applied a threshold of 0.500, as suggested by Pappas and Woodside (2021). For Chinese firms, we followed Greckhamer (2016) and set the PRI threshold at 0.650.

5. Results

5.1. Necessity analysis

We conducted a necessity analysis to check whether any single factor is essential for achieving high organizational resilience. The results suggest that no individual factor qualifies as a necessary condition. Specifically, using the *superSubset* function from the QCA package to assess necessary relationships in kind (Vis & Dul, 2018), we found that all conditions had consistency values below the 0.900 threshold (Ragin, 2008), except for the absence of innovating in U.S. firms (0.916). This result likely reflects the low prevalence of this strategy in the U.S. sample rather than a substantive necessity. Following prior studies (e.g., Mattke et al., 2022), we do not interpret it as a meaningful necessary condition.

Table 3

Descriptive statistics and calibration threshold values for U.S. and Chinese firms.

Conditions/Outcome	Calibration threshold values			Measure descriptives			
	Fully in	Crossover	Fully out	Mean	SD	Max	Min
U.S. firms (N = 717)							
Crisis perception	—	—	—	0.682	0.466	1.000	0.000
Persevering	—	—	—	0.840	0.367	1.000	0.000
Retrenchment	—	—	—	0.750	0.433	1.000	0.000
CSR practices	—	—	—	0.506	0.500	1.000	0.000
Innovating	—	—	—	0.078	0.269	1.000	0.000
Strategic flexibility	−0.599	0.000	0.559	−0.022	0.349	0.764	−1.178
Organizational change capacity	3.761	0.000	−2.978	0.165	2.018	10.908	−5.444
Organizational resilience	0.274	0.000	−0.486	−0.015	0.266	1.565	−0.996
Chinese firms (N = 890)							
Crisis perception	—	—	—	0.796	0.404	1.000	0.000
Persevering	—	—	—	0.851	0.357	1.000	0.000
Retrenchment	—	—	—	0.512	0.500	1.000	0.000
CSR practices	—	—	—	0.537	0.499	1.000	0.000
Innovating	—	—	—	0.475	0.500	1.000	0.000
Strategic flexibility	−0.357	0.000	0.485	0.021	0.258	0.871	−0.696
Organizational change capacity	3.125	0.000	−2.345	0.154	1.691	8.110	−4.365
Organizational resilience	0.110	0.000	−0.104	0.002	0.070	0.364	−0.236

5.2. Sufficiency analysis (main analysis)

Table 4 shows the results of the sufficiency analysis. The consistency scores for each configuration and solution formula are above the recommended threshold of 0.750 (Ragin, 2008), indicating the validity of our analysis.

As shown in Table 4, three configurations are sufficient to achieve high resilience for either U.S. firms (US1-US3) or Chinese firms (CN1-CN3). Configuration US1 suggests that U.S. firms can sustain high resilience through crisis perception and persevering strategies despite lacking strategic flexibility, change capacity, retrenchment, and CSR

Table 4

Configurations sufficient for achieving high resilience for U.S. and Chinese firms (main analysis).

Conditions	Configurations for U.S. firms			Configurations for Chinese firms		
	US1	US2	US3	CN1	CN2	CN3
Crisis perception	●	●	●	●	●	●
Crisis response strategies						
Persevering	●	●	●	●	●	●
Retrenchment	⊗	●	●	⊗	⊗	⊗
CSR practices	⊗	⊗	●	⊗	●	●
Innovating		⊗	●	⊗	⊗	●
Dynamic capabilities						
Strategic flexibility	⊗	●	⊗	⊗	●	●
Organizational change capacity	⊗	⊗	●	●	⊗	●
Consistency	0.893	0.908	0.887	0.872	0.847	0.893
PRI score	0.623	0.555	0.687	0.693	0.670	0.689
Raw coverage	0.041	0.087	0.021	0.033	0.063	0.034
Unique coverage	0.041	0.087	0.021	0.033	0.063	0.034
Solution consistency, PRI score, coverage	0.901, 0.604, 0.149			0.865, 0.680, 0.130		

Notes: Full circles indicate the presence or high level of a condition. Crossed-out circles indicate the absence or low level of a condition. Blank spaces indicate that a condition may be either absent or present. Large circles represent the core conditions. Small circles represent the peripheral conditions.

practices. The core condition of crisis perception enables firms to effectively identify threats and opportunities, allowing them to maintain operations with persevering strategies despite insufficient organizational capability.

Configuration US2 indicates that U.S. firms possessing strong strategic flexibility yet limited change capacity can develop high resilience by combining crisis perception, persevering, and retrenchment while avoiding CSR practices and innovating. The core conditions of crisis perception, retrenchment, and strategic flexibility enable firms to reallocate resources and maintain core operations in response to environmental changes.

Configuration US3 demonstrates that U.S. firms with high change capacity but restricted strategic flexibility can build high resilience through a comprehensive approach incorporating crisis perception, persevering, retrenchment, CSR practices, and innovating. With innovating as the core condition, these firms leverage their strong resource capacity to implement multiple strategies simultaneously, compensating for limited flexibility through innovative solutions.

Configuration CN1 reveals that Chinese firms with change capacity can maintain high resilience through crisis perception and persevering strategies, while lacking other conditions. The core condition of crisis perception, combined with strong resource capacity, enables firms to weather the crisis solely through persevering strategies.

Configuration CN2 shows that Chinese firms with strong strategic flexibility but inadequate change capacity can obtain high resilience by engaging in crisis perception, persevering, and CSR practices while avoiding retrenchment and innovating. The core conditions of CSR practices and strategic flexibility reflect these firms' emphasis on flexible resource allocation to maintain stakeholder relationships for collective crisis management.

Configuration CN3 demonstrates that Chinese firms can achieve high resilience by adopting all practices except retrenchment. The multiple core conditions (CSR practices, innovating, strategic flexibility, and change capacity) indicate these firms' ability to fully utilize their resources and flexibility advantages while maintaining strong stakeholder relationships and implementing innovating strategies to manage uncertainty.

In summary, both U.S. and Chinese firms developed an "operations maintenance" configuration emphasizing crisis perception and persevering strategies. However, U.S. firms additionally formed "resource retrenchment" and "innovative transformation" configurations based on either strategic flexibility or change capacity. Chinese firms, according to their combinations of these two dynamic capabilities, explored "CSR-

oriented” and “CSR-innovation integration” configurations.

5.3. Comparative analysis of configurations for U.S. And Chinese firms

Comparing the configurations for U.S. and Chinese firms in Table 4, the following differences can be observed.

Retrenchment appears in two U.S. configurations (US2 and US3) but is absent from all Chinese configurations. This contrast may reflect the U.S. regulatory environment’s emphasis on employment-at-will doctrines and shareholder primacy, which legitimizes cost-cutting measures during crises. In China, formal employment protection policies and informal expectations of maintaining social stability likely constrained firms’ retrenchment options, especially in the early pandemic phase.

CSR practices emerge as core conditions in Chinese configurations CN2 and CN3, while appearing only as a peripheral condition in US3. This pattern may demonstrate how China’s state-led governance model explicitly channels corporate resources toward social objectives during crises, reinforced by collectivist values prioritizing the public good. U.S. firms, operating in a context that tends to separate business and social responsibilities, approach CSR through a market-driven, voluntary lens.

Dynamic capabilities are present in all Chinese configurations, with some featuring multiple core capabilities—a pattern less evident in U.S. configurations. This could reveal China’s institutional environment characterized by frequent policy shifts and government intervention, requiring firms to develop superior adaptive capabilities. In the U.S., by contrast, regulatory stability and easier access to mature capital markets may reduce firms’ dependence on internal dynamic capabilities.

Overall, these differences underscore the influence of institutional contexts: firms in market-led environments tend to rely more on market-based mechanisms like retrenchment, while being less dependent on dynamic capabilities. Conversely, firms in state-led environments show crisis responses more strongly shaped by institutional demands and social norms, with a stronger emphasis on CSR and dynamic capabilities.

5.4. Dynamic analysis

To examine how resilience configurations evolve over time, we collected two longitudinal datasets from 2020 to 2022 and employed panel fsQCA (Garcia-Castro & Ariño, 2016). Data on crisis perception and response strategies were derived from keyword matching in periodic reports closest to June 30 and December 31 of each year, resulting in six periods: 2020Q2, 2020Q4, through 2022Q4. Organizational resilience was measured as the industry-adjusted ratio of a firm’s ORPPC during each period relative to its pre-crisis level in December 2019. Dynamic capabilities were measured using a five-year moving window approach and lagged by one year. Calibration procedures followed those in the main analysis. We set the consistency threshold at 0.800, the PRI threshold at 0.500, and adjusted frequency thresholds (38 and 26) to retain approximately 80 % of cases.

Table 5 presents the results, including two configurations (US4 and US5) for U.S. firms and three configurations (CN4 to CN6) for Chinese firms, each sufficient for resilience from 2020 to 2022.

Based on the progression of COVID-19, we define 2020Q2 as the short-term and 2020Q2 to 2022Q4 as the long-term of the crisis. Comparing short-term configurations (Table 4) with long-term ones (Table 5) reveals a temporal shift: crisis perception and persevering decline over time, while retrenchment, CSR, and innovating become more prominent, reflecting a shift from initial vigilance and stability to long-term adaptation. Specific discussions follow.

First, the influence of crisis perception weakens over time. While it appears in all short-term configurations, it becomes less prominent in the long term, suggesting that heightened perception was especially vital during the initial phase of uncertainty.

Second, the effect of persevering diminishes. Present in all short-term configurations, it becomes more marginal in the long term. This reflects

Table 5

Configurations sufficient for achieving high resilience for U.S. and Chinese firms (dynamic analysis).

Conditions	Configurations for U.S. firms		Configurations for Chinese firms		
	US4	US5	CN4	CN5	CN6
Crisis perception	●	⊗	●	⊗	⊗
Crisis response strategies					
Persevering	•	•	⊗	⊗	•
Retrenchment	⊗	●			
CSR practices	⊗	●		●	
Innovating					●
Dynamic capabilities					
Strategic flexibility		⊗	•	●	•
Organizational change capacity	⊗		●	⊗	•
Consistency	0.779	0.766	0.803	0.821	0.793
PRI score	0.479	0.414	0.522	0.477	0.484
Raw coverage	0.026	0.073	0.036	0.020	0.019
Unique coverage	0.026	0.073	0.036	0.020	0.019
BECONS distance	0.010	0.008	0.019	0.011	0.029
WICONS distance	0.003	0.006	0.005	0.004	0.004
Solution consistency, PRI score, coverage	0.769, 0.432, 0.099		0.805, 0.501, 0.075		

Notes: Full circles indicate the presence or high level of a condition. Crossed-out circles indicate the absence or low level of a condition. Blank spaces indicate that a condition may be either absent or present. Large circles represent the core conditions. Small circles represent the peripheral condition.

that merely maintaining operations was initially crucial, but gradually gave way to more proactive strategies.

Third, dynamic capabilities exhibit divergent trends across contexts. U.S. firms increasingly move away from relying on dynamic capabilities (present in US2 and US3 but absent in Table 5), while Chinese firms maintain them throughout. This may reflect institutional differences: stabilizing COVID-19 measures in the U.S. versus China’s dynamic zero-COVID policy, which necessitated sustained adaptive capabilities among Chinese firms.

Fourth, retrenchment, CSR practices, and innovating become more prominent over time, consistently appearing as core conditions in long-term configurations. This highlights the enduring importance of sustainable practices and strategic adjustments in building long-term competitive advantages throughout the crisis period.

We next assess the temporal stability of each configuration using three metrics from Garcia-Castro and Ariño (2016): (1) between cross-sectional measures of consistency (BECONS), measuring consistency in each period; (2) BECONS distance, reflecting variation across periods; and (3) pooled consistency (POCONS), indicating overall consistency across the entire timeframe.

As shown in Table 5, all configurations have BECONS distances below the benchmark (0.065, $T = 6$) from Garcia-Castro and Ariño (2016), suggesting relative stability. Fig. 2 shows that BECONS for each configuration decreased during the mid-to-late stages of COVID-19, followed by a slight recovery. This decline is likely due to firms adapting to the new normal as the pandemic stabilized, with a slight increase in consistency in 2022 due to challenges such as the Omicron variant. Chinese firms exhibit greater BECONS volatility, likely due to the continuous adjustments required by the government’s dynamic zero-COVID policy. These findings suggest that the effectiveness of resilience configurations is influenced by both crisis evolution and institutional stability.

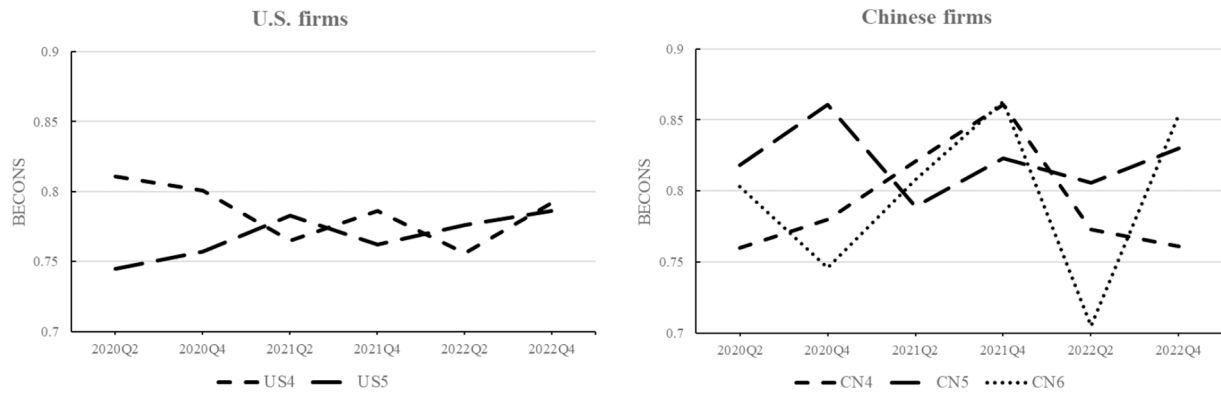


Fig. 2. BECONS for each configuration from 2020 to 2022.

6. Discussion

This study investigates how organizations achieve resilience through different configurations of dynamic capabilities and crisis responses across institutional and temporal contexts during COVID-19. Using fsQCA on U.S. and Chinese manufacturing firms, we identify distinct patterns leading to resilience. Table 6 summarizes three key findings: (1) short-term configurations that enable resilience in both U.S. and Chinese firms, (2) institutional differences between market-led and state-led environments, and (3) the temporal evolution of configurations as the

crisis progressed. These findings show that organizational resilience arises from complex alignments between capabilities, strategies, and institutional contexts and evolves over time.

6.1. Theoretical contributions

Our study contributes to the literature on organizational resilience in three ways. First, we propose an integrated perspective that combines the capability and process views of organizational resilience. Previous research has often approached organizational resilience from either a

Table 6
Summary of key findings of this study.

Analytical dimension	Configuration characteristics	Core conditions	Key findings	Theoretical/Practical implications
Short-term configurations of U.S. firms	• US1: operations maintenance	• Crisis perception	• Achieving resilience through crisis perception and persevering despite lacking strategic flexibility, change capacity, retrenchment, and CSR	• Demonstrates how firms maintain resilience with limited resources and capabilities
	• US2: resource retrenchment	• Crisis perception, retrenchment, strategic flexibility	• Effectively reallocating resources and maintaining core operations through strategic flexibility despite limited change capacity	• Illustrates market-oriented resource optimization
	• US3: innovative transformation	• Innovating	• Driving resilience through innovation-led comprehensive strategies	• Highlights the critical role of innovation in resilience development
Short-term configurations of Chinese firms	• CN1: operations maintenance	• Crisis perception	• Maintaining resilience through crisis perception and persevering with change capacity	• Emphasizes the importance of basic resource capacity
	• CN2: CSR-oriented	• CSR practices, strategic flexibility	• Emphasizing flexible resource allocation and stakeholder relationships	• Reflects a collective crisis management approach
	• CN3: CSR-innovation integration	• CSR practices, innovating, strategic flexibility, change capacity	• Fully utilizing resources and flexibility advantages to innovate while maintaining stakeholder relationships	• Shows a comprehensive resilience-building approach
Cross-institutional comparison	• Retrenchment patterns	—	• Present in U.S. configurations (US2, US3) but absent in Chinese configurations	• Highlights preference for retrenchment under market-driven institutional logics
	• CSR implementation	—	• Core in Chinese configurations (CN2, CN3); peripheral in U.S. (US3)	• Indicates stronger institutional and normative pressures for CSR in state-led environments
	• Dynamic capabilities	—	• More prevalent in Chinese configurations	• Highlights state intervention in China elevates firms' need for dynamic capabilities
Temporal evolution	• Crisis perception	—	• Diminishing effect over time	• Shows the dynamic nature of crisis cognition
	• Strategic changes	—	• Weakening role of persevering strategies	• Reveals strategic adaptation patterns
	• Dynamic capabilities	—	• Increasing importance of retrenchment, CSR, and innovation	• Indicates paths to long-term competitive advantage
			• Decreasing reliance in U.S. firms	• Shows how policy volatility in China vs. regulatory stability in the U.S. shapes firms' long-term reliance on dynamic capabilities
			• Sustained emphasis in Chinese firms	

capability view, focusing on organizational capabilities, or a process view, emphasizing crisis responses (Do et al., 2022; Duchek, 2020; Hillmann & Guenther, 2021; Khan et al., 2024), resulting in a fragmented understanding of how these antecedents interact. Drawing on the concept of strategic fit (Venkatraman, 1989; Zajac et al., 2000), we posit that organizational resilience emerges from the effective alignment between organizational capabilities and crisis responses. Using fsQCA, we identify various configurations that illustrate this alignment to achieve high resilience. For instance, configuration US2 shows that organizations with high strategic flexibility may benefit more from resource-efficient actions like retrenchment, while configuration US3 indicates that firms with strong change capacity could better leverage resource-intensive measures such as innovating. By identifying these nuanced configurations, our study offers a deeper understanding of how capabilities and responses align to create organizational resilience.

Second, by comparing the differences in configurations that lead to high resilience between U.S. and Chinese firms, we reveal how specific institutional features shape capability-response configurations. While prior research has largely focused on single-country settings (Napier et al., 2024; Su & Junge, 2023), our study responds to recent calls to incorporate comparative institutional perspectives into resilience research (Ciravegna & Michailova, 2022). Based on the concept of strategic fit, we argue that organizational resilience depends not only on the alignment between crisis responses and internal capabilities but also on their matching with the institutional environment. Our findings show that U.S. firms—operating in a context of regulatory stability and individualist culture—are more likely to adopt market-based mechanisms such as retrenchment and view CSR as discretionary. In contrast, Chinese firms—embedded in a state-led framework with stronger government intervention and collectivist culture—tend to emphasize CSR and dynamic capabilities in their crisis responses. These patterns highlight the importance of both formal and informal institutions in shaping how firms configure their resilience strategies.

Third, by employing the dynamic fsQCA method, we provide more nuanced insights into the temporal evolution of organizational resilience. While existing research emphasizes the dynamic nature of organizational resilience (Conz & Magnani, 2020; Jiang et al., 2021), few studies have empirically examined how the relationships between resilience and its antecedents change over time. By utilizing panel fsQCA with two longitudinal datasets, we find that the configurations through which organizations leverage dynamic capabilities and crisis responses to build resilience differ between the short-term and long-term phases of a crisis. For example, crisis perception and persevering are more prominent in the short term, suggesting that initial crisis periods involve higher environmental uncertainty and a greater need to maintain core operations. Furthermore, we find that the BECONS of configurations leading to high resilience changes over time, potentially due to varying crisis severity across different periods. These findings contribute to the literature by uncovering differences in the antecedent configurations of organizational resilience across timeframes in both their components and associational strength.

6.2. Managerial implications

This study provides important implications for businesses, especially large manufacturing firms, in crisis management. In market-led environments like the U.S., where regulatory stability and mature market mechanisms reduce uncertainty, dynamic capabilities are relatively less critical. Firms in such contexts should prioritize crisis perception and early warning systems to maintain operations. Those with strategic flexibility can reallocate resources to streamline operations and control costs, while firms with strong change capacity can pursue innovation-driven strategies. For instance, Ashland, a large U.S. chemical company, leveraged strategic flexibility during the early stages of COVID-19 to retrench and adjust its business model and cost structure, ensuring operational continuity.

By contrast, state-led environments like China involve more government intervention and policy volatility, requiring firms to develop stronger dynamic capabilities to navigate formal institutional demands. Meanwhile, informal norms rooted in collectivist culture emphasize social responsibility. Firms with strong change capacity can implement emergency measures and adapt quickly, while those with strategic flexibility should manage stakeholder relations and reallocate resources to support collective interests. When both capabilities are present, firms can more effectively integrate CSR and innovation. For example, Shanghai Challenge Textile Co., Ltd. utilized both capabilities early in the pandemic to shift production to isolation gowns in response to government directives and expand into health protection markets, achieving growth amid crisis.

In the short term of a crisis, firms should focus on immediate crisis perception and perseverance strategies to ensure basic operations. As the crisis persists, firms may need to adopt more sustainable measures, such as retrenchment, CSR, and innovation, to establish long-term competitive advantages. For instance, Tsinghua Tongfang Co., Ltd., a leading Chinese electronics firm, ensured business continuity in early 2020 by focusing on core operations. As the pandemic continued, it seized market opportunities, expanding into online work and epidemic prevention testing, achieving a recovery in the later stages of the crisis.

These findings also offer implications for public policy. To foster organizational resilience, policymakers should maintain a stable yet adaptive institutional environment, for example, avoiding both excessive intervention and regulatory neglect. Short-term measures such as targeted subsidies or tax relief can help resource-constrained firms reallocate resources and respond effectively. In the long term, governments should promote strategic transformation by supporting digital upgrading or green transition initiatives, aligning with firms' innovation and CSR strategies to build sustainable advantages.

6.3. Limitations

Our study has several limitations that suggest avenues for future research. First, while we developed a concise framework to explain the complex antecedents of organizational resilience, it does not capture all influential factors. As COVID-19 has accelerated digital transformation (Browder et al., 2024), future research could incorporate digital capabilities into the framework. Moreover, crisis responses extend beyond crisis perception and the four strategies examined in our study; organizational reflection and learning, for instance, play crucial roles in resilience (Williams et al., 2017). Future research could explore how these factors align to enhance resilience.

Second, our empirical approach has limitations, particularly in the binary coding of crisis responses. Future research could explore hybrid methods (e.g., weighted keyword counting) to capture response intensity. Additionally, we adopted a 1 %/99 % exclusion rule to eliminate extreme data fluctuations, which may overlook meaningful outliers. Future studies should develop more granular exclusion criteria based on firm-specific evidence or conduct in-depth analyses of these extreme cases, especially in small sample contexts.

Finally, the applicability of our framework may be bounded by specific contextual conditions. Our sample focuses on large manufacturing firms, whose resource base and operational characteristics differ substantially from those of small and medium-sized enterprises (SMEs) or service-oriented firms. For instance, SMEs may face greater challenges in maintaining operational continuity, while service firms may rely more heavily on innovation-based responses (Li et al., 2022). Furthermore, our analysis focuses exclusively on the COVID-19 pandemic. Although this crisis provides a valuable context for studying resilience, different types of disruptions—such as natural disasters, geopolitical conflicts, or financial shocks—may require alternative combinations of capabilities and response strategies. Future research should test the framework across various crisis types and organizational contexts to better understand its boundary conditions and

generalizability.

CRediT authorship contribution statement

Chao Zhou: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Yuan Huang:** Writing – review & editing, Funding acquisition, Conceptualization. **Renhuai Liu:** Writing – review & editing,

Validation, Supervision, Resources, Conceptualization.

Acknowledgments

This study was supported by the Humanities and Social Sciences of the Ministry of Education Planning Program of China (24YJA630032) and by the Natural Science Foundation of Sichuan Province of China (2024NSFSC1045).

Appendix

This appendix includes a robustness analysis of the findings from the main text and a list of all industries selected in the study sample.

Robustness analysis.

Robustness checks were performed to examine the stability of the findings. First, we tested higher (0.820) and lower (0.780) consistency thresholds in both the main and dynamic analyses. The higher threshold excluded CN4 and CN6 in the dynamic analysis for Chinese firms, while the lower threshold introduced two additional configurations similar to CN1 and CN3. Other results remained unchanged. We also lowered the frequency thresholds by one (main analysis) and two cases (dynamic analysis), which led to only minor changes in consistency and coverage.

Second, we constructed alternative measures of organizational resilience by using ORPPC data from the following quarter as the numerator in calculating the resilience ratio (e.g., Q3 2020 instead of Q2 2020). This adjustment yielded one additional configuration similar to US5 for U.S. firms and one new configuration for Chinese firms in the main analysis; in the dynamic analysis, CN5 was replaced by a new configuration. Despite these variations, our main conclusions remained consistent.

Finally, to address the sample size disparity between U.S. and Chinese firms, we conducted stratified sampling by industry to achieve comparable sample sizes. This resulted in the exclusion of CN3 and CN4 from the main and dynamic analyses for Chinese firms, respectively, while all other findings remained stable. These results suggest that sample size differences do not substantially affect our key conclusions.

Table A1
List of the industries included in the study.

SIC code	Name of the industry for U.S. firms	CSRC code	Name of the industry for Chinese firms
20	Food and kindred products	C13	Agricultural and sideline food processing industry
22	Textile mill products	C14	Food manufacturing
23	Apparel and other finished products	C15	Alcohol, beverage, and refined tea manufacturing
25	Furniture and fixtures	C17	Textile industry
26	Paper and allied products	C18	Textile garment and apparel industry
27	Printing, publishing, and allied industries	C19	Leathers, furs, feathers, and related products and footwear industry
28	Chemicals and allied products	C21	Furniture manufacturing
30	Rubber and miscellaneous plastic products	C22	Papermaking and paper product industry
31	Leather and leather products	C23	Printing and recording media reproduction industry
32	Stone, clay, glass, and concrete products	C24	Manufacturing of stationery, industrial arts, sports, and entertainment supplies
34	Fabricated metal products, except machinery and transportation equipment	C26	Manufacturing of chemical raw materials and chemical products
35	Industrial and commercial machinery and computer equipment	C27	Pharmaceutical industry
36	Electronic and other electrical equipment and components, except computer equipment	C28	Chemical fiber manufacturing
371	Motor vehicles and motor vehicle equipment	C29	Industry of rubber and plastic products
375	Motorcycles, bicycles, and parts	C30	Industry of non-metallic mineral products
387	Watches, clocks, clockwork-operated devices, and parts	C33	Metal product industry
391	Jewelry, silverware, and plated ware	C34	General equipment manufacturing
393	Musical instruments	C35	Special-purpose equipment manufacturing
394	Dolls, toys, games, and sporting and athletic	C36	Automobile manufacturing
395	Pens, pencils, and other artists materials	C375	Motorcycle manufacturing
396	Costume jewelry, costume novelties, buttons, and miscellaneous notions, except precious metal	C376	Manufacturing of bicycles and disabled vehicles
399	Miscellaneous manufacturing industries	C38	Electric machinery and equipment manufacturing
		C39	Manufacturing of computers, communications, and other electronic equipment
		C403	Manufacturing of watches and timing instruments
		C41	Other manufacturing industries

Data availability

Data will be made available on request.

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